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REVIEW

# Development and applications of deep eutectic solvents in different chromatographic techniques

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## Abstract

Deep eutectic solvents (DESs) being inexpensive, easy to prepare, biocompatible, biodegradable and environmentally benign have aroused as the most propitious greener solvents for designing novel analytical methods of chemical analysis specially covering almost major fields of chromatography. This review article is majorly concerned with the analytical evaluation of DESs as new generation solvents for use as (a) modifier of silica gel surface via silylation reaction to create new stationary phases, (b) mobile phase (or developer) in liquid chromatography and (c) mobile phase additive to improve chromatographic performance. The sequential evolution of DESs has been briefly presented in tabular form since their beginning from the development of choline chloride/urea eutectic mixture in 2003. It has been observed that three more important properties (ionic conductivity, polarity and viscosity) of DESs can be adjusted to the desired level by proper selection of molar composition of hydrogen bond donor and hydrogen bond acceptor and water content. In this review, we present the recent applications of DESs in chromatography covering the period of the last two decades for assessing their suitability as a new class of green solvents.

**Keywords** Deep eutectic solvents · Supercritical fluid chromatography · Counter-current chromatography · Gas chromatography · Capillary electrochromatography · Natural deep eutectic solvents (NADES)

## 1 Introduction

Organic solvents are volatile and produce a lot of waste, which would be detrimental to the environment and human health. Organic solvents are being replaced by green or less toxic solvents, for example, ILs (ionic liquids) and DESs (deep eutectic solvents) in separation science particularly in chromatography, in the twenty-first century, based on twelve principles of green chemistry. *Abbott* et al. [1] were the first to prepare the deep eutectic phenomenon in the year of 2003. These workers blended urea and choline chloride having melting points of 133 °C and 302 °C, respectively, in the molar ratio of 1:2 to obtain an eutectic mixture (melting point: 12 °C) that was liquid at room temperature. Because of alike physical and chemical properties to ILs, DESs were classed as sub-group of ILs. Furthermore, DESs are preferred over ILs as solvents because of their ease of production, biocompatibility, biodegradability, cost-effective starting materials and low toxicity. Because one of the components is of natural origin, *Choi* et al. [2] coined the term natural deep eutectic solvents (NADES) for DESs. Amino acids, sugars, organic acids, terpenes and phenolic chemicals are just a few examples of primary metabolites used to prepare NADES. Since the first report, NADES have been used as an extraction medium particularly for polyphenolics. Table 1 shows the evolution of DESs throughout the history [3]. The commonly used hydrogen bond donors (HBDs) and hydrogen bond acceptors (HBAs) taken together in various molar ratios for the synthesis of numerous eutectic mixtures are shown in Fig. 1a and b. According to the literature survey, only a few review articles have appeared on the applications of DESs in chromatography [4–6]. *Lenca* and coworkers have published an admirable review in 2017 on the applications of room-temperature ionic liquids in liquid chromatography [7].

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**Table 1** Summary of chronological development of deep eutectic solvents (DESs) [3]

| Discovery | Year | Pathfinder |
|---|---|---|
| Development of intermixed solvent systems composed of quaternary ammonium salts and metal chlorides contributed in big way to the nascent field of DESs | 2001 | A.P. Abbott |
| Liquid-phase systems consisting of quaternary ammonium salts and urea in defined molar ratios at ambient temperature were coined as DESs | 2003 | A.P. Abbott |
| Exploration of more mixed solvents based on choline chloride and carboxylic acids or hydrated metal salts as novel DESs | 2004 | A.P. Abbott |
| Formation of ambient temperature DESs through appropriate combinations of metal salts ($ZnCl_2$) and certain donors (amides and diols) | 2007 | J.C. Barron |
| Uncovering of the natural deep eutectic solvents (NADES) | 2011 | Y.H. Choi |
| Designing of hydrophobic DESs based on quaternary ammonium salts and decanoic acid (DA) | 2015 | Van Osch |
| Introduction of supramolecular mixtures consisting of methyl-β-cyclodextrine (RAMEB) and levulinic acid called SUPRADES | 2020 | Tracy El Achkar |

According to the literature review, DESs have been widely employed in different analytical applications including separation techniques. A review on the application of green solvents in thin-layer chromatography (TLC) was recently published by *Qasim Ullah* et al. [8]. Furthermore, in recent years, DESs have received more attention of researchers, resulting in the arising number of studies appearing in the literature about properties [9–12] and their applications [13–21]. *Abbott* and coworkers [22] categorized DESs into four groups based on their compositional components as shown in Fig. 2.

According to the literature survey, type III DESs analogs to ILs in terms of physicochemical properties have been commonly investigated in DES system. A mixture of thymol (HBAs) and menthol (HBDs) has recently been recognized as DES and categorized as type V [23]. Cyclic oligosaccharides (CD) which are non-toxic and obtained from the enzymatic degradation of polysaccharide (starch) were used as HBAs for the production of supramolecular mixtures in the liquid state at room temperature [24, 25]. Furthermore, based on water solubility, DESs have been classified as i) hydrophilic deep eutectic solvents and ii) hydrophobic deep eutectic solvents [26]. The replacement of traditional solvents by DESs as reaction media in synthetic organic [27, 28] and enzymatic reactions [29, 30] has encouraged researchers working in different fields of chemical analyses including preparative chromatography for successful applications of DES. [Tables 7, 8, 9].

## 2 Solvent polarity

Some unique characteristics of DESs including their ability of solubilizing compounds with different polarities, the capability to tune polarity as per requirement and efficacy of improving solubility of selected compounds have encouraged researchers to use eutectic solvents as alluring alternatives to traditional organic solvents. The polarity of a solvent is an important parameter which plays an important role in separation science, specifically in chromatographic separation techniques. Semi-empirical linear free energy relationships have been used to characterize the polarity of solvents. The ultraviolet−visible (UV−Vis) absorbance spectrophotometric methods based on betaine dye [31–33], Kamlet−Taft (KT) [34–37] and Nile red polarity parameters have been widely used to explain the structural effects of DESs on the magnitude of overall polarity [38, 39]. The various probes that have been extensively used to measure solute−solvent interactions and the polarity of DESs include Reichardt's betaine dye 30 and 33, DENA, NA, Py, PyCHO, pyridine-N-oxide, Nile red, prodan, C153, TNS and ANS. The values of different solvatochromic parameters measured for various DESs (binary and ternary systems) using these scales since the first report appearing in 2011 [38] have been encapsulated in Tables 2, 3, 4. Further, only a few papers have appeared in the literature describing the use of fluorescence-dependent probes in the determination of polarity [40, 41]. The empirical scale of solvent 'polarity, $E_T(30)$, is defined as the molar transition energy of the dye in kcal per mole at room temperature and normal pressure by using the following equation: $E_T(30) = 28{,}591/\lambda_{max}$ in nm.

$E_T(33)$ defined as the lowest energy absorbance transition of betaine dye 33 was determined using the following relationship:

$$E_T(30) = 28591/\lambda_{max}, \quad (1)$$

$E_T(33)$ measured in kcal per mole at $\lambda_{max}$ (wavelength corresponding to maximum absorption of probe) was used as the indicator of the solvent's overall polarity, dipolarity/polarizability and /or HBD ability, arising from interactions between the solvent and the dye.

The Kamlet and Taft scale, $\alpha$, $\beta$ and $\pi^*$ are three main parameters where α represents hydrogen bond-donating

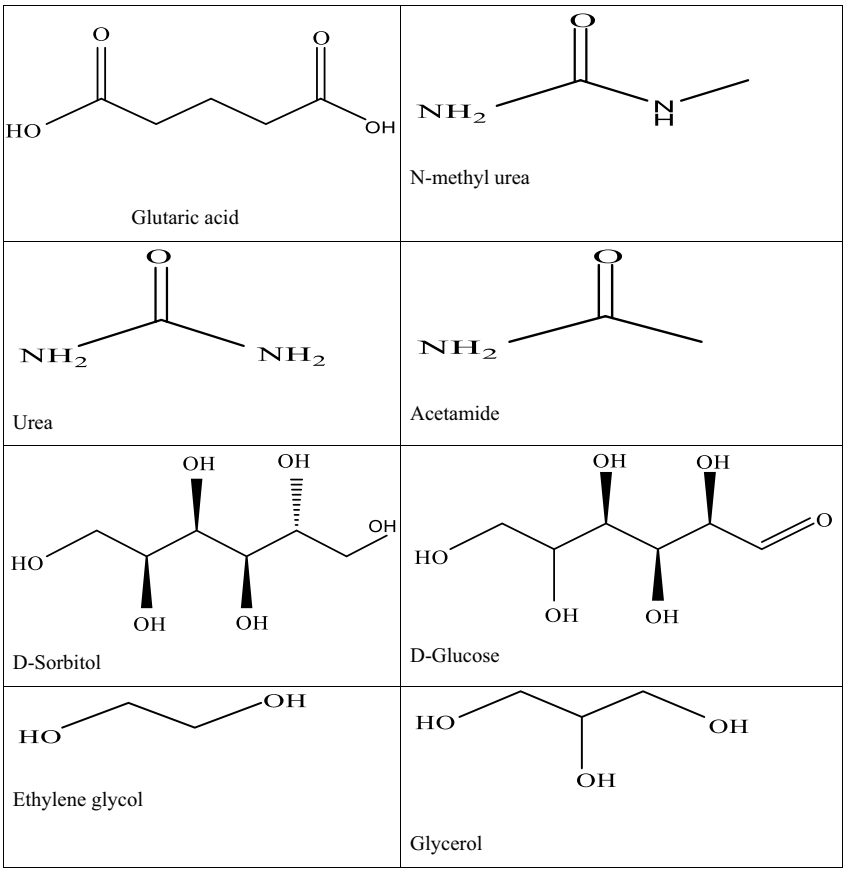



**Fig. 1** The structures of commonly used HBDs (**a**). The structures of commonly used HBAs (**b**)

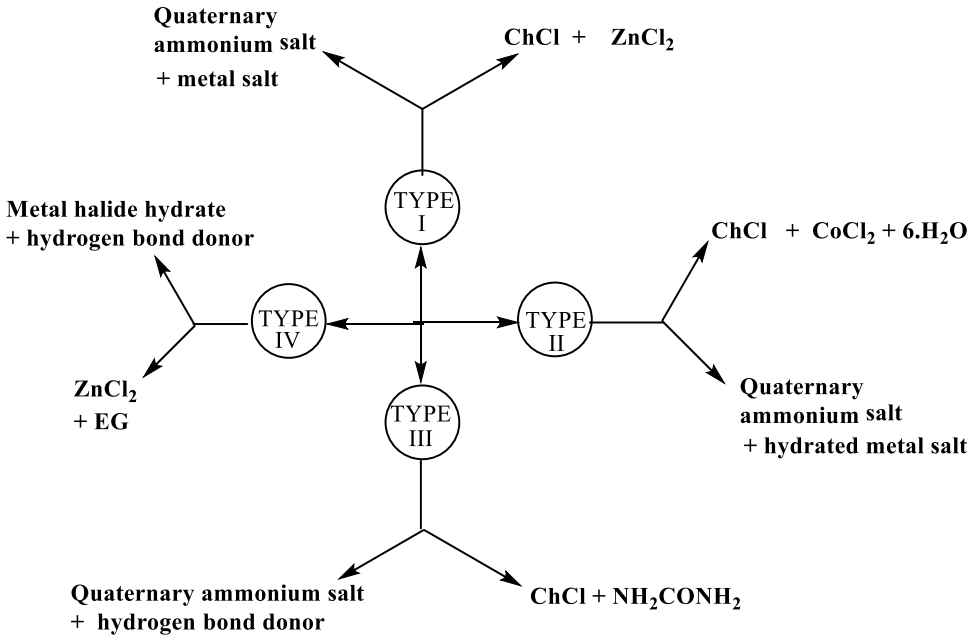



**Fig. 2** Classification of DESs on the basis of their constituents

acidity (HBD), i.e., the proton-donating capacity of the solvent. It can be calculated using the following relationship:

$$\alpha = 0.0649E_{\mathrm{T}}(33) - 2.03 - 0.72\pi^{*}. \quad (2)$$

Here $E_{\mathrm{T}}$ is calculated using Eq. (1) whereas $\pi^{*}$ is calculated using the following equation:

$$\pi^{*} = 0.314(27.52 - \mathrm{v}_{N,N\text{ - dimethyl - 4 - nitroaniline}}) \quad (3)$$

Here $v$ is the experimental wave number, i.e., $v$ ($\mathrm{cm}^{-1}$) = 107 428/λmax (nm).

The $\beta$ parameter represents hydrogen-bonding basicity, i.e., the proton-accepting ability of the solvent which generally depends upon the type of the anionic part of HBA [42]. The value of $\beta$ can be calculated by using the following equation:

$$\beta = \left(1.035\mathrm{v}_{N,N\text{ - dimethyl - 4 - nitroaniline}} + 2.64 - v_{4\text{–nitroaniline}}\right)/2.80. \quad (4)$$

The $\pi^{*}$ parameter represents the dipolarity/polarizability of a solvent arising due to non-specific interactions (dipole – dipole and dipole-induced dipole interactions) as well as the polarizability operating between the solute (dye) and solvents (DESs). The value of $\pi^{*}$ can be calculated by using Eq. (3).

Nile red is a very important probe, widely used to estimate the polarity of various DESs [43]. In this case, the shift in absorption maxima was measured and ENR was calculated using the following equation:

$$\mathrm{ENR} = 28591/\lambda_{\max}. \quad (5)$$

In contrast to betaine dye 30, Nile red showed bathochromic shift in polar solvents and a hypsochromic shift in non-polar solvents [44]. It was found that ENR values were inversely proportional to the polarity of the substance, i.e., the higher ENR values obtained from Nile red probe, the lower was the polarity of the substance and vice versa. Certain favourable properties (photochemical stability, adequate wider solubility in solvents), and low basicity have made Nile red a popular probe to estimate the polarity of various DESs. Owing to the higher sensitivity and orthogonality of information, fluorescence techniques are being considered as important tools to estimate the polarity of various DESs. Pyrene has been the most commonly used probe in this technique. However, only limited papers have been published on the use of fluorescence-based probes to measure solvent polarity [40, 41, 45].

## 3 Methods for preparing DESs

For preparation of DESs, the following methods are generally employed.

**Table 2** Summary of the reported Kamlet−Taft parameters (α, β and π*) of binary DESs systems

| DES composition | | | Kamlet−Taft parameters | | | References |
|---|---|---|---|---|---|---|
| HBD | HBA | HBD:HBA | α | β | π* | |
| Glycerol (Gly) | – | – | 0.882 | 0.658 | 0.956 | [11] |
| Glycerol (Gly) | ChCl | 1:1 | 0.923 | 0.658 | 1.003 | [11] |
| Glycerol (Gly) | ChCl | 1.5:1 | 0.921 | 0.658 | 0.980 | [11] |
| Glycerol (Gly) | ChCl | 2:1 | 0.937 | 0.657 | 0.984 | [11] |
| Glycerol (Gly) | ChCl | 3:1 | 0.914 | 0.657 | 0.970 | [11] |
| Glycerol (Gly) | ChCl | 3:1 | 0.914 | 0.657 | 0.970 | [11] |
| Butanoic acid (BA) | $[N_{2222}]Cl$ | 2:1 | 0.99 | 0.76 | 0.92 | [12] |
| Hexanoic acid (HA) | | | 0.97 | 0.85 | 0.86 | [12] |
| Octanoic acid (OA) | | | 0.96 | 0.87 | 0.81 | [12] |
| Butyric acid (BA) | $[N_{3333}]Cl$ | 2:1 | 0.94 | 0.84 | 0.93 | [12] |
| Hexanoic acid (HA) | | | 0.91 | 0.92 | 0.85 | [12] |
| Octanoic acid (OA) | | | 0.90 | 0.96 | 0.80 | [12] |
| Butyric acid (BA) | $[N_{4444}]Cl$ | 2:1 | 0.92 | 0.99 | 0.86 | [12] |
| Hexanoic acid (HA) | | | 0.90 | 1.02 | 0.81 | [12] |
| Octanoic acid (OA) | | | 0.84 | 1.19 | 0.80 | [12] |
| Decanoic acid (DA) | $[N_{4444}]Cl$ | 1:1 | 0.91 | 1.21 | 0.86 | [12] |
| Butyric acid (BA) | $[N_{3333}]Br$ | 2:1 | 1.07 | 0.80 | 0.93 | [12] |
| Hexanoic acid (HA) | | | 1.02 | 0.86 | 0.87 | [12] |
| Octanoic acid (OA) | $[N_{4444}]Br$ | 2:1 | 1.02 | 0.81 | 0.93 | [12] |
| Hexanoic acid (HA) | | 1:1 | 1.09 | 0.84 | 0.90 | [12] |
| Octanoic acid (OA) | | 1:2 | 0.94 | 0.82 | 0.95 | [12] |
| Decanoic acid (DA) | | 2:1 | 1.02 | 0.93 | 0.92 | [12] |
| | | 2:1 | 0.98 | 1.09 | 0.84 | [12] |
| | | 2:1 | 0.95 | 1.05 | 0.71 | [12] |
| Butyric acid (BA) | ChCl | 2:1 | 0.945 | 0.775 | 0.514 | [13] |
| Butyric acid (BA) | | 3:1 | 0.914 | 0.847 | 0.526 | [13] |
| Pantanoic acid (PA) | | 2:1 | 0.830 | 0.703 | 0.495 | [13] |
| Pantanoic acid (PA) | | 3:1 | 0.785 | 0.752 | 0.557 | [13] |
| Decanoic acid (DA) | | 2:1 | 0.421 | 0.667 | 0.853 | [13] |
| Decanoic acid (DA) | | 3:1 | 0.327 | 0.620 | 0.955 | [13] |

i. *Heating method:* This method involves mixing and heating of HBDs and HBAs at approximately 373 K with constant stirring till a clear homogeneous liquid is obtained which requires about 1 to 1.5 h [46].
ii. *Grinding method:* In this process, the components are mixed at room temperature and pulverized till a homogenous mixture in the liquid form is obtained [47].
iii. *Evaporation method:* This method entails dissolving of starting materials in water using rotary device, followed by evaporation at 323 K under vacuum to remove the water. The resulting liquid is stored in a desiccator containing silica gel until it reaches a constant mass [48].
iv. *Freeze-drying method:* This process involves combining both the HBD and HBA components in the presence of about 5% of water. These mixed solutions are frozen, and then freeze-dried to produce a clear homogenous liquid [49]. In comparison to the previous three ways, this method is not frequently used in the preparation of DESs.
v. *Microwave radiation method:* Due to intrinsic advantages such as greater yields, softer reaction conditions and shorter reaction durations, this method is the most promising for the manufacture of NADES [50].
vi. *Microwave-assisted method: Gomez* et al. [51] developed a microwave-assisted irradiation process for the

**Table 3** Summary of the reported Kamlet−Taft parameters ($\alpha$, $\beta$ and $\pi^*$) of ternary DESs systems

| DES composition | | | Kamlet−Taft parameters | | | References |
|---|---|---|---|---|---|---|
| HBD | HBA | HBD:HBA | $\alpha$ | $\beta$ | $\pi^*$ | |
| Urea<br>Glycerol (Gly) | ChCl | 1:1:1 | 0.893 | 0.583 | 1.170 | [14] |
| Urea<br>Ethylene glycol (EG) | | | 0.862 | 0.560 | 1.165 | [14] |
| Urea<br>Formamide (FA) | | | 0.846 | 0.593 | 1.159 | [14] |
| Urea<br>Thiourea | ChCl | 1:1:1 | 0.840 | 0.486 | 1.245 | [14] |
| Glycerol (Gly)<br>Ethylene glycol (EG) | | | 0.897 | 0.625 | 1.127 | [14] |
| Glycerol (Gly)<br>Formamide | ChCl | | 0.891 | 0.598 | 1.131 | [14] |
| Glycerol (Gly)<br>Thiourea | | | 0.889 | 0.539 | 1.217 | [14] |
| Ethylene glycol (EG) | ChCl | 1:1:1 | 0.864 | 0.601 | 1.207 | [14] |
| Formamide<br>Thiourea | | | 0.843 | 0.549 | 1.209 | [14] |

**Table 4** The polarity values for binary and ternary DESs as measured using betaine dye 30 and betaine dye 33

| DES composition | | | $E_T(30)$ in kcal per mol | $E_T(33)$ in kcal per mol | $E_T^N$ in kcal per mol | References |
|---|---|---|---|---|---|---|
| HBD | HBA | HBD:HBA | | | | |
| Glycerol (Gly) | ChCl | 1:1 | 58.49 | – | 0.858 | [11] |
| | | 1.5:1 | 58.21 | – | 0.849 | |
| | | 2:1 | 58.58 | – | 0.860 | |
| | | 3:1 | 57.96 | – | 0.841 | |
| Urea<br>Glycerol (Gly) | ChCl | 1:1:1 | – | – | 0.843 | [14] |
| Urea<br>Ethylene glycol (Gly) | ChCl | 1:1:1 | – | – | 0.827 | [14] |
| Urea<br>Formamide | ChCl | 1:1:1 | – | – | 0.817 | [14] |
| Urea<br>Thiourea | ChCl | 1:1:1 | – | – | 0.844 | [14] |
| Glycerol (Gly)<br>Ethylene glycol (Gly) | ChCl | 1:1:1 | – | – | 0.830 | [14] |
| Glycerol (Gly)<br>Formamide | ChCl | 1:1:1 | – | – | 0.829 | [14] |
| Glycerol (Gly)<br>Thiourea | ChCl | 1:1:1 | – | – | 0.857 | [14] |
| Ethylene glycol (Gly)<br>Formamide | ChCl | 1:1:1 | – | – | 0.814 | [14] |
| Ethylene glycol (Gly)<br>Thiourea | ChCl | 1:1:1 | – | | 0.842 | [14] |
| Formamide<br>Thiourea | ChCl | 1:1:1 | – | – | 0.833 | [14] |

synthesis of DESs that is eco-friendly, takes less time and uses less energy.

vii. *Ultrasound-assisted synthesis:* Ultrasonic waves can also be used to prepare DES [52, 53]. This approach produces DESs that are stable throughout time and do not crystallize even after several days [54]. Furthermore, physicochemical parameters such as density, viscosity, and decomposition temperature as well as Fourier-transform infrared spectroscopy (FTIR) spectra were comparable to those DESs made using other procedures such as heating and stirring methods [55].

## 4 Applications of DESs in different chromatographic technologies

Since a long time, organic volatile solvents have been utilized in many types of chromatographic techniques. However, the high cost, enormous consumption and waste generation have a negative impact on human health and even cause pollution in the environment. As a result, these volatile solvents have been replaced by environmentally benign green solvents such as water, supercritical fluid (SCF), ILs and DESs. DESs have been recommended as attractive and highly performing green solvent systems in terms of sensitivity, selectivity and precision for researchers in almost all types of chromatographic methods as a mobile phase additive/component, stationary phase impregnant/modifier and stationary phase materials.

In fact, the DESs-containing mobile phases have shown improved separation efficacy. However, several issues which need to be resolved before using DESs in LC include: (i) high viscous nature which restricts the easy flow of mobile phase through stationary phase material packed in column or spread over TLC plate, (ii) strong inter-linking with stationary phases which affects separation performance efficiencies leading to the shortening of column life, (iii) lack of availability of the high-performance liquid chromatography (HPLC) grade DESs and (iv) low volatility of DESs which can have negative effects on detector systems. Due to these drawbacks, DESs have not been proven to be as suitable mobile phase solvents as they deserve for chromatography. Natural DESs, on the other hand, have been discovered to be possible optional solvents in liquid chromatography [56]. They focused on future research and concluded that the development of relevant technologies is required before natural DESs may be utilized in HPLC analysis. These advancements include high-temperature endurable instrument, an improved pump system to provide high back pressure and stationary phases that allow chromatographic analysis at a high temperature to reduce DESs viscosity.

### 4.1 Gas chromatography (GC)

*Boczkaj* et al. [57] first employed DESs as stationary phases in gas chromatography. The stationary phase comprised of a mixture of heptadecanoic acid (HDC) as HBD (melting point of 61.3 °C) and tetrabutylammonium chloride (TBAC) as HBA (melting point of 70 °C) in 2:1 molar ratio and was found to be appropriate for the resolution of volatile organic compounds (VOC). The selectivity efficacy of DES-based synthesized stationary phase for large number of organic compounds (alcohols, ketones, thiophenes and aromatic hydrocarbons) was determined and compared with the selectivity pattern of commercially available GC stationary phases utilizing the values of the Rohrschneider–McReynolds constants [58–61] as shown in Tables 5 and 6, respectively. The Rohrschneider–McReynolds constants indicate that the synthesized DES was the stationary phase of medium polarity. However, the newly synthesized stationary phase with DES has strong interactions with alcohols, because of (i) intermolecular hydrogen bonding between the DES component (HBD and HBA) and the –OH group of alcohols, and (ii) hydrophobic interactions of the DES with the alkyl chains of alcohols, while retention increases significantly with carbon chain length. In comparison to other alcohols, *n*-hexyl alcohol and *n*-heptyl alcohol were found to have higher retention. As a result, the retention periods of these two alcohols are by 191% and 300% longer, respectively, than the expected values based on their boiling points. The findings of this study expand the range of applications for DESs by allowing them to be used as sorptive media in chromatographic procedures. The properties of DESs can be modified by either changing one of its components or adding solute dissolved in DES attributed to a lot of possibilities for developing new stationary phases. The retention data presented in Table 5 suggest that these might be applied for other separation techniques (e.g., extractive distillation).

**Table 5** Retention data of organic volatile compounds measured for newly developed DES stationary phases comprising of heptadecanoic acid (HBD) and tetrabultylammonium chloride (TBAC) as HBA in 2:1 molar ratio [58–61]

| Compound | M.W. (g/mol) | B.P. (°C) | $t_r$ (Mint) | $R_F$ ($k$) | Theoretical $t_r$ by B.P. (Mint) | $\Delta t_r$ (%) | Ip |
|---|---|---|---|---|---|---|---|
| Aromatic hydrocarbons | | | | | | | |
| Benzene | 78.1 | 80.1 | 3.49 | 7.3 | 2.09 | 66.64 | |
| Methylbenzene | 92.14 | 110.6 | 5.48 | 12.0 | 3.78 | 45.07 | 51 |
| Ethylbenzene | 106.17 | 136 | 7.24 | 16.2 | 5.75 | 25.98 | 40 |
| Propylbenzene | 120.19 | 159 | 9.45 | 21.5 | 7.98 | 18.49 | 24 |
| Butylbenzene | 134.22 | 183 | 13.99 | 32.3 | 10.75 | 30.11 | 27 |
| 1,3-Dimethylbenzene (o-Xylene) | 106.17 | 144 | 8.16 | 18.4 | 6.47 | 26.04 | 31 |
| Alcohols | | | | | | | |
| *n*-Propyl alcohol | 60.6 | 97 | 6.85 | 15.3 | 2.94 | 133.39 | 106 |
| *n*-Butyl alcohol | 74.12 | 116 | 9.25 | 21.0 | 4.15 | 122.74 | 101 |
| *n*-Pentyl alcohol | 88.15 | 138 | 13.7 | 31.7 | 5.92 | 131.77 | 90 |
| *n*-Hexyl alcohol | 102.17 | 156 | 22.3 | 52.1 | 7.66 | 191.09 | 79 |
| *n*-Heptyl alcohol | 116.2 | 175 | 39.3 | 92.7 | 9.78 | 302.43 | 90 |
| *sec*-Butyl alcohol | 74.12 | 98 | 6.7 | 15.0 | 2.99 | 24.61 | 87 |
| *sec*-Pentyl alcohol | 88.15 | 119 | 8.7 | 19.7 | 4.37 | 99.24 | 70 |
| *tert*-Amyl alcohol | 88.15 | 102 | 6.68 | 14.9 | 3.23 | 106.97 | 58 |
| Cyclohexyl alcohol | 116.2 | 160 | 15.4 | 35.8 | 8.08 | 91.04 | 65 |
| Ketone | | | | | | | |
| Cyclopentanone | 84.12 | 131 | 7.07 | 15.8 | 5.32 | 32.93 | 71 |
| Cyclohexanone | 98.14 | 155.6 | 9.96 | 22.7 | 7.62 | 30.72 | 48 |
| 3-Methylcyclohexanone | 112.17 | 169 | 12.26 | 28.2 | 9.08 | 35.07 | 43 |
| 2-Pentanone | 86.13 | 101 | 4.38 | 9.4 | 3.17 | 38.28 | 45 |
| 2-Hexanone | 100.16 | 127.6 | 6.41 | 14.3 | 5.04 | 27.22 | 27 |
| 3-Heptanone | 114.19 | 147 | 8.1 | 18.3 | 6.76 | 19.82 | 23 |
| Methylisopropyl ketone | 86.13 | 101 | 3.7 | 7.8 | 2.66 | 38.97 | 38 |
| Diisopropyl ketone | 114.19 | 124 | 5.8 | 12.7 | 4.75 | 21.21 | 8 |
| Acetyl acetone | 100.12 | 140 | 7.1 | 15.9 | 6.10 | 16.48 | 31 |
| Sulfide | | | | | | | |
| Diethyl sulfide | 90.19 | 91 | 3.61 | 7.6 | 2.61 | 38.29 | 27 |
| Dipropyl sulfide | 122.25 | 142 | 7.24 | 16.2 | 6.29 | 15.15 | 10 |
| Dibutyl sulfide | 146.29 | 188 | 14.17 | 32.7 | 11.39 | 24.42 | 16 |
| Dimethyl disulfide | 94.2 | 110 | 5.27 | 11.5 | 3.74 | 41.02 | 33 |
| Diethyl disulfide | 122.25 | 152 | 8.84 | 20.0 | 7.25 | 21.89 | 23 |
| Dipropyl disulfide | 150.31 | 193 | 17.93 | 41.7 | 12.05 | 48.85 | 22 |
| Di-t-butyl disulfide | 178.36 | 200 | 17.57 | 40.0 | 13.35 | 31.61 | -21 |
| Thiophene and its alkylated derivatives | | | | | | | |
| Thiophene | 84.14 | 84 | 3.99 | 8.5 | 2.27 | 75.92 | 46 |
| 2-Methylthiophene | 98.17 | 112 | 5.74 | 12.7 | 3.87 | 48.22 | 27 |
| 3-Methylthiophene | 98.17 | 116 | 6.07 | 13.5 | 4.15 | 46.16 | 29 |
| 2-Ethylthiophene | 112.19 | 133 | 7.5 | 16.9 | 5.49 | 36.68 | 22 |
| Other compounds | | | | | | | |
| Pyridine | 79.1 | 115.2 | 8.5 | 19 | 4.10 | 108.02 | 91 |
| Nitropropane | 89.09 | 132 | 6.9 | 15.4 | 5.40 | 27.53 | 57 |

**Table 6** Comparison of McReynolds constants values for developed DES phase and commercial stationary phases [58–61]

| Phase | Benzene<br>$X'$<br>$\Delta I$ | 1-Butanol<br>$Y'$ | 2-Pentanol<br>$Z'$ | Nitropropane<br>$U'$ | Pyridine<br>$S'$ | Sum<br>$E(\Delta I)$ |
|---|---|---|---|---|---|---|
| DES phase | 57 | 395 | 147 | 278 | 298 | 1174 |
| DES phase (repeated) | 56 | 396 | 147 | 279 | 299 | 1177 |
| SPB-50 | 125 | 175 | 183 | 268 | 220 | 971 |
| Span 80 | 97 | 266 | 170 | 216 | 268 | 1017 |
| OV-1 | 16 | 55 | 44 | 65 | 42 | 222 |
| OV-25 | 178 | 204 | 208 | 305 | 280 | 1175 |
| Diethoxyethyl sebacate | 151 | 306 | 211 | 320 | 274 | 1262 |
| DC QF-1 (FS 1265) | 144 | 233 | 355 | 463 | 305 | 1500 |
| Atper 282 | 200 | 108 | 186 | 235 | 289 | 1100 |
| Dibutoxyethyl phthalate | 151 | 282 | 227 | 338 | 267 | 1265 |
| Carbowax 20 M | 322 | 536 | 368 | 572 | 510 | 2308 |
| SPB-50 | 125 | 175 | 183 | 268 | 220 | 971 |

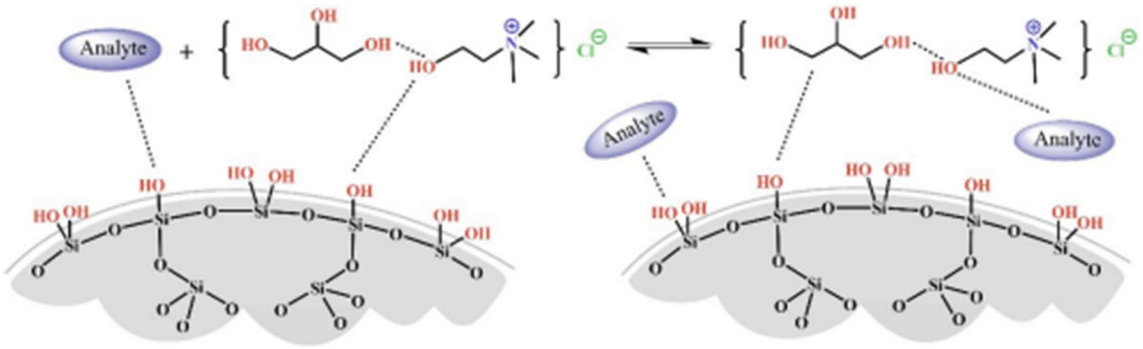



**Fig. 3** Mechanistic for ChCl–Gly (1:2 molar ratio) as mobile phase additive [62]

## 4.2 Supercritical fluid chromatography (SFC)

*Ping* et al. [62] reported the first paper on the use of DESs as a novel mobile phase additive in SFC for the simultaneous resolution and identification of ten alkaloids such as coptisine (COP), berberine (BER), stepharanine (STE), tetrahydropalmatine (THE), jatrorrhizine (JAT), magnoflorine (MAG), phellodendrin (TED) and sinomenine (SIN) belonging to the isoquinoline family, abbreviated as IQAs. They used supercritical $CO_2$ and methanol as mobile phase A, whereas mobile phase B comprised of a mixture of 2% $H_2O$, 0.5% formic acid (FA), and 0.25% choline chloride−glycerol (ChCl−Gly) (1:2 molar ratio) and a Zorbax RX-SIL column filled with hydroxylated silica surface acted as the stationary phase. The addition of DES composed of ChCl−Gly (0.25%) to the mobile phase as an additive was shown to significantly

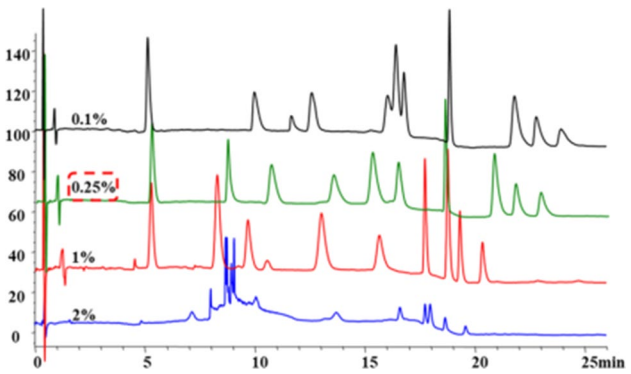



**Fig. 4** Effect of concentrations of DES (ChCl–Gly) as mobile phase on the separation of IQAs [62]

reduce silanol effect, improve alkaloids resolution, and minimize peak tailing. Moreover, DES acted as silanol blocker to reduce the concentration level of active silanol group sites and thus competes with IQAs for adsorption on silica surface as shown in Fig. 3. Additionally, the proposed "green" method was found to be the most suitable for the analysis of alkaloids present in medicinal herbs (MH).

The following parameters mainly affect the resolution of IQAs as discussed below.

i. *Nature of stationary phase:* This is the most important parameter which mainly impacts the chromatographoic behaviour of IQAs in SFC analysis. Different stationary phases like Agilent Zorbax (SB-C18, RX-SIL and Eclipse XDB-C8) columns along with Thermo Scientific Syncronis Amino column were used to optimize the peak shapes and resolution of IQAs [63]. Among these, Zorbax RX-SIL column gave satisfactory results with good peak shapes for the IQAs compared to the other reversed-phase columns. It was probably due to the high content of silica (≥ 99.99%) in the column material which resulted in reducing peak tailing via reduction in the silanol effect. Moreover, the separation of basic compounds like IQAs with different structures was difficult on the fully hydroxylated column and other reversed-phase columns.
ii. *Mobile phase additives:* Mobile phase additives play an important role in reducing the magnitude of active silanol groups on the silica surface because the modifier competed with the analytes for HBD/HBA sites on stationary phase surface [64]. Separation of IQAs with methanol containing mixed solvents in the absence of DES was not possible. However, improved separations of all IQAs were found when 0.25% ChCl−Gly (1:2) was used as the modifier. As a result, the mobile phase composed of MeOH–2% $H_2O$–0.5% FA–0.25% ChCl−Gly was found the best for the better resolution of IQAs compared to other mobile phase systems. It happened due to the combined effect of HBDs and HBAs [65]. The hydrogen bonding among free silanol groups on the silica surface and DESs attributed to the improved resolution of IQAs.
iii. *DES concentration:* The retention duration for all IQAs was decreased with increase in the concentration of ChCl−Gly from 0.1% to 1% in the mobile phase as shown in Fig. 4.
iv. *Back pressure, temperature and flow rate*

*Jiang* et al. [66] investigated the effect of temperature variation on density and viscosity of the supercritical $CO_2$ fluid

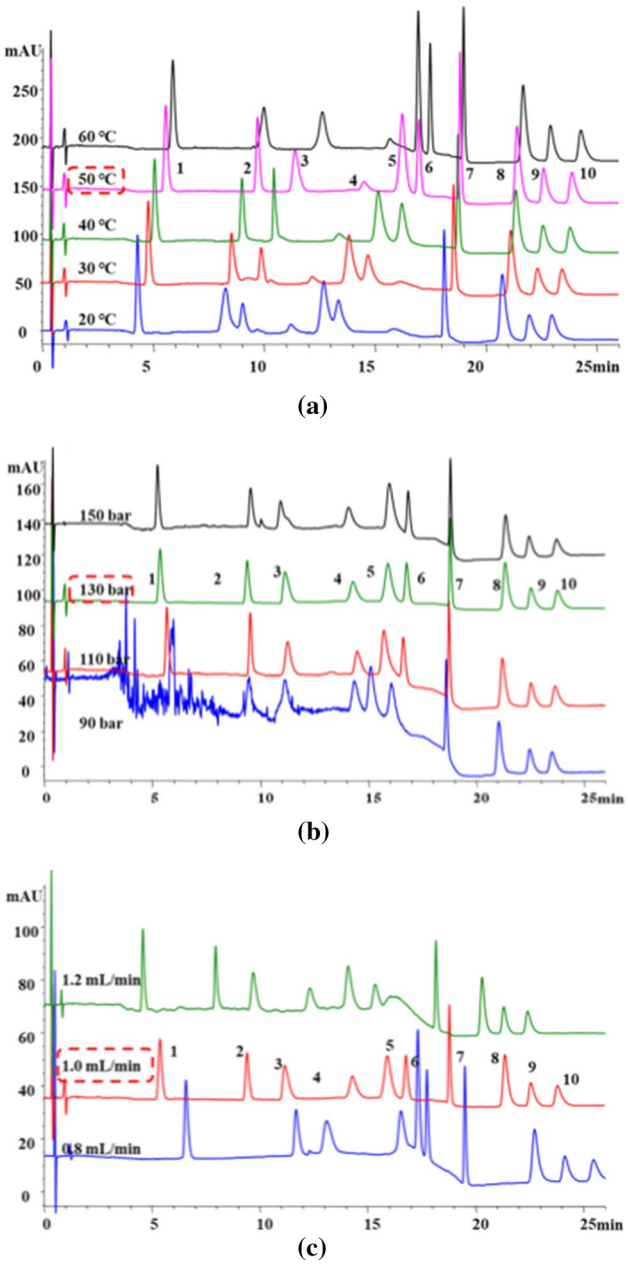



**Fig. 5** Effect of temperature on the performance of SFC (**a**). Effect of back pressure on the performance of SFC (**b**). Effect of flow rate on the performance of SFC (**c**) [62]

which affects the retention of the IQAs. Back pressure has been another important parameter in the separation of IQAs. Figure 5a–c also demonstrate the effect of back pressure, temperature and flow rate on the separation of all IQAs. *Ping* et al. [62] studied the effects of back pressure and temperature on the separation of IQAs. They observed that with

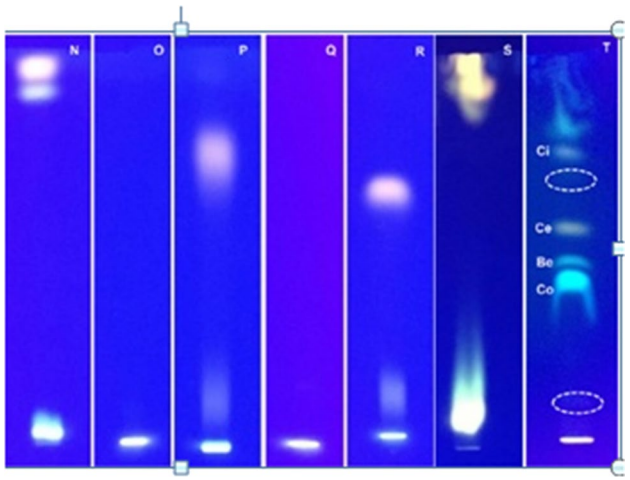



**Fig. 6** Chromatograms obtained with the selected mobile phases, using Si60 HPTLC plates under exposer UV radiation (366 nm) developed with pure solvents: *N* methanol, *O* water, *P* acetone; *Q* chloroform, *R* diethyl ether, *S* the working solution developed in D4-M60, *T C. maius* root extract developed with D4-M35 mobile phases. Alkaloids marked in the chromatograms: *Be* berberine, *Ce* chelerythrine, *Ci* chelidonine, *Co* coptisine, *Sa* sanguinarine [68]

increase in column temperature, both density and viscosity of the mobile phase were reduced, which attributed to the decrease in retention of compounds and deteriorated the resolution between compound pairs SIN and BER and JAT and STE. Thus, 50 °C was found the most appropriate column temperature [67].

The flow rate of supercritical carbon dioxide played a decisive role on the separation of analytes [63]. The influence of different flow rates (0.8–1.2 mL/min) on the retention sequence of IQAs was examined where an inverse relationship between the flow rate of the mobile phase and retention of IQAs was observed.

### 4.3 Thin-layer chromatography (TLC)

The use of DESs in TLC was first reported by *Raj* in the year 2020 [68]. In all, she prepared 25 eutectic mixed systems during her experiment using camphor, phenol, choline chloride or proline as HBA and phenyl salicylate, citric acid or malic acid as HBD, mostly in 1:1 molar ratio of HBA/HBD. Most of the systems were classified as NADES. However, DESs based on terpenes and phenols were new which have not been previously used for analytical purposes. All these systems were used either as pure solvents or after dilution with water, acetone, chloroform, methanol, or diethyl ether during the experiment. Among these eutectic mixtures, *Raj* used equimolar mixture of menthol and phenol with addition of 35% methanol as the most effective mobile phase for resolving all five examined isoquinoline

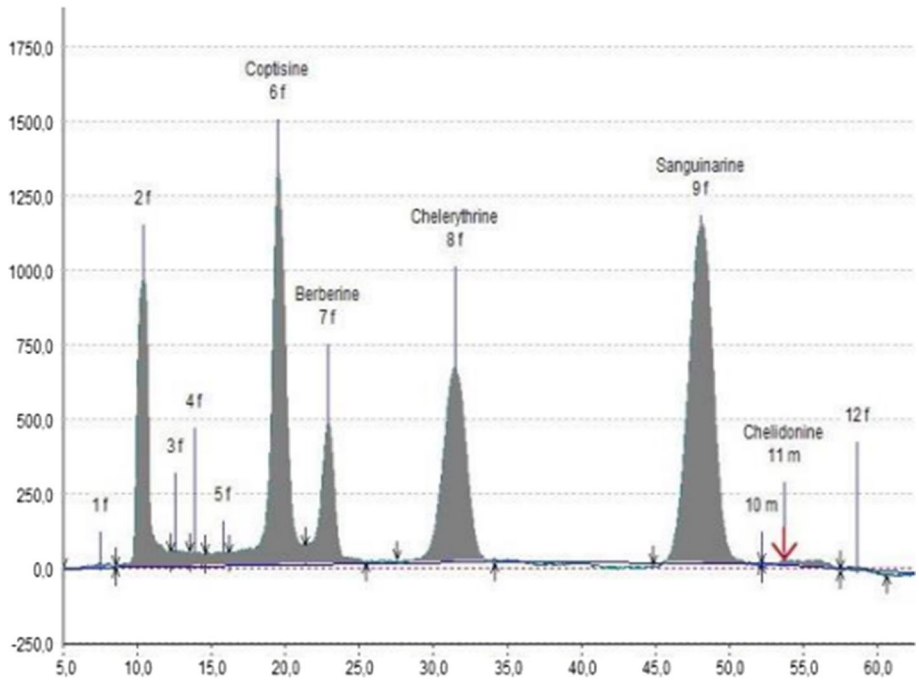



**Fig. 7** Densitogram of resolution of real sample of alkaloids obtained at 340 nm using LiChrospher HPTLC plates with EM3–M20 mobile phase [69]

**Table 7** Summary of DESs applications as mobile phase/mobile phase modifier in chemical analysis of various groups of compounds using different chromatographic techniques

| Analytes/sample matrix | Experimental conditions | Remarks/reference |
|---|---|---|
| Alkaloids (berberin, chelerythrine, chelidonine, copticine and sanguinarine) / *C. maius* plant root extract | SP: Si60 HPTLC plates,<br>MP: A mixture of methanol and DES in 35:65 (*w/w*) ratio,<br>DES composition: menthol + phenol in 1:1 molar ratio<br>AT: HPTLC−densitometry with detection under UV radiation | In all 264 different eutectic liquids (numbers given in parenthesis) in the pure form (25) and after dilution (*w/w*) with traditional solvents like acetone (20), methanol (104), chloroform (20), diethyl ether(20) or water (100) were designed as DESs systems for analysis of alkaloids by TLC. Under optimized experimental conditions, all five alkaloids present in *C. maius* were successfully resolved on silica HPTLC plates [68] |
| Alkaloids (berberin, chelerythrine, chelidonine, copticine and sanguinarine) / *C. maius* plant root extract | SP: Silica gel (Si60) HPTLC plates<br>MP: A mixture of DES with 20% added methanol in 2:1 (*w/w*) ratio<br>DES composition: Camphor + phenol (1:1) and menthol + lemonine (1:1)<br>AT: HPTLC−densitometry, detection under UV light | This is the first paper on the use of quantitative eutectic TLC with DES as one of the components of mobile phase [69] |
| Caffeic acid/standard mixture | SP: $C_{18}$ column<br>MP: A mixture of 0.10% DES in methanol and water in 18:82 ratios by volume<br>DES composition: Gly: ChCl in 3:1 molar ratio,<br>HPLC−UV−VIS | Under the optimum experimental conditions, DES as additive shortened the retention time, modified the chromatogram shape and enhanced the column efficiency in HPLC. This method was found superior than that in which phosphoric acid was used as additive [70] |
| Quercetin/standard mixture | SP: $C_{18}$ column<br>MP: A mixture of DES (0.20%) + methanol + water in 60:40 ratios by volume<br>DES composition: EG:ChCl in 2:1 molar ratio,<br>AT: HPLC−UV−VIS | Under the optimum conditions, the proposed mobile phase reduced the tailing factor to the level of 1.71 for 6690 theoretical plates. This method was found most suitable for the determination of bioactive compounds with the use of HPLC [71] |
| Protocatechuic acid derivatives/standard solution | SP: $C_{18}$ column<br>MP: A mixture of DES and 5% ethyl alcohol in 95:5 ratios by volume<br>DES composition: lactic acid−glucose−water in 5:1:4 molar ratio,<br>AT: HPLC−UV | Three DESs with different compositions as one of the component of mobile phase were examined, out of those, the most effective mobile phase was a mixture of DES and 5% ethyl alcohol in 95:5 ratios by volume for the resolution of protocatechuic acid derivatives [72] |
| Quaternary alkaloids (sanguinarine, berberine chloride, coptisine chloride and chelerythrine)/Herbal oral solution | SP: $C_{18}$ column<br>MP: A mixture of acetonitrile (ACN) + 1% aqueous solution of DES in 32:68 ratios by volume (pH 3.3)<br>DES composition: EG:ChCl in 3:1 molar ratio,<br>AT: HPLC−UV | This is the first report on the use of DES in RP-HPLC to achieve improved separation of bioactive quaternary alkaloids (QAs). Among seven examined DESs systems, 1% DES as one of the components in mobile phase was found effective to decrease the peak tailing attributed and improving resolution of 4 QAs. The limits of detection (LODs) for 4 QAs were in the range of $6 \times 10^{-3}$ to $2 \times 10^{-3}$ µg/mL [73] |
| Cardiovascular drugs: metformin (MTF), metoprolol (MTP) atorvastatin (ATV) and aspirin (ASA) /Urine, human plasma | SP: $C_{18}$ column<br>MP: A mixture of SDS (0.09 M) + 10% butanol + 3.5% NADES + 3.5% glacial acetic acid by volume<br>DES composition: EG:ChCl in 2:1 molar ratio,<br>AT: MLC−UV | The proposed micellar liquid chromatography(MLC) was used to separate and quantify the combination of 4 cardiovascular drugs in the samples of urine and human plasma. Under the optimum conditions, the drugs were effectively separated within a 12 mints when the flow of mobile phase was 1 $cm^3$/min. The limits of detection were in the range of 0.03–25 mg/L [74] |

**Table 7** (continued)

| Analytes/sample matrix | Experimental conditions | Remarks/reference |
|---|---|---|
| Melamine (MEL)/Milk | SP: $C_{18}$ column<br>MP: A mixture of 92% SDS (0.10 M) + 4% NADES + 4% glacial acetic acid by volume<br>DES composition: EG:ChCl in 2:1 molar ratio,<br>AT: MLC–DAD | This MLC system was found effective for monitoring the MEL level in milk samples under optimized experimental conditions through reduced peak broadening. MEL was eluted within 10 min without interference from coexisting proteins present in milk. The limit of detection was 0.075 mg/L and the reported method was found promising for analysis of other polar ionizable compounds [75] |

alkaloids (chelerythrine, coptisine, sanguinarine, berberine and chelidonine) on Si60 high-performance thin-layer chromatography (HPTLC) plates. All these alkaloids existing in the genus *Chelidonium* plant were detected using UV light (366 nm) with preliminary separation on TLC plates (Fig. 6). The developed TLC system was successful to separate alkaloids present in the real *Chelidonium maius* extract samples.

Though the proposed method is simple for the separation and identification of alkaloids without derivatization, the low volatility of the mobile phase has been problematic for using spraying reagents as detectors. In fact, the residual mobile phase penetrating inside silica gel particles hinder the visualization of the detected spots on TLC plates with clarity after spraying detecting reagents. Furthermore, the longer development time (3–4 h) can be considered another demerit of this proposed method in spite of the fact that the diluted DES migrates at faster rate than the pure DES.

As an extension of her previous work published in 2020 as the first publication, *Raj* [69] introduced the term "Eutectic thin-layer chromatography" for her work on the quantitative analysis of five alkaloids existing together in *C. maius*. In this method, she used Si60 plates as the stationary phase and a mixture of DESs (camphor + phenol and menthol + limonene in 2:1 ratio by weight) with addition of 20% methanol for the quantitative estimation of berberine, capticine, sanguinarine, chelidonine and chelerythrine after extraction from dried plant material (Fig. 7). The developed TLC method was validated and tested on the real sample and found operational for qualitative as well as quantitative analysis.

As per literature survey, DESs have been used as mobile phases only for the analysis of alkaloids by TLC using silica gel as the stationary phase. It is therefore desirable to broaden the scope of TLC use in combination of DESs as mobile phases for analysis of other groups of organic as well as inorganic compounds. To speed up the process, DESs with low viscosity have to be developed. Furthermore, the use of reversed-phases (bonded hydrophobic phases) as substitutes for hydrophilic silica gel and the two-dimensional mode of TLC may be beneficial which need to be investigated.

### 4.4 Counter-current chromatography and centrifugal chromatography

The applications of DESs in chromatography (listed in Table 7) as per references [68–75] are related to the use of DESs as single homogenous phases. To widen the applicability of DESs, *Garcia* and co-workers [76] first introduced the use of DES-based non-aqueous biphasic systems for the separation of hydrophobic compounds by counter-current chromatography/centrifugal partition chromatography

**Table 8** Applications of DESs containing bi-phasic mobile phase systems in chemical analysis by liquid−liquid chromatography (CPC and CCC)

| Analytes/Sample matrix | Experimental conditions | Remarks/Reference |
|---|---|---|
| α-Tocopherol, β-ionone, retinol and carvone/ Standard mixtures | MP: A mixture of ethanol, heptane and DES in 40:30:30 ratio by weight,<br>DES composition: levulinic acid−ChCl in 2:1 molar ratio,<br>AT: Liquid−liquid chromatography (CCC and CPC) | Application of developed DES-based biphasic solvent system in separation of multi-component mixtures of hydrophobic compounds by liquid−liquid chromatography (LLC) using centrifugal partition extractor (CPE) connected with a pair of isocratic HPLC pumps. As stationary phase, DES-rich liquid system was used in the ascending mode (flow rate, 40 mL/min) but as mobile phase it was used in descending flow form (flow rate, 20−40 mL/min). As neoteric water-free biphasic solvent, DES−ethanol−heptane system was found highly reliable to resolve the mixtures of naringenin, ionone, retinol, carvone and tocopherol because the partition coefficients of these compounds fall in the preferred range (0.4−2.5) of LLC, performed in a centrifugal partition extractor (CPE) [76] |
| Tocopherol homologues/Standard mixtures | MP: A mixture of heptane, ethanol and CBD (ChCl + 1,4-butanediol) in 30:40:30 weight ratio,<br>DES composition: 1,4-butanediol (BD)−ChCl in 1:1 molar ratio<br>AT: CPC−UV | The predictive thermodynamic model COSMO-RS was used to identify the most favourable DES-based biphasic water-free solvent combinations for selective separations of tocopherol homologues by CPC. The well resolved peaks with separation factor of 1.7 were realized for the separation of α-tocopherol from γ-tocopherol. The proposed computational solvent screening selection approach has enormous promising potential in chemical analysis [77] |
| Coumarin, β-ionone and α-tocopherol/Standard mixtures | MP: A mixture of methanol, *n*-heptane and DES in various weight ratios,<br>DES composition: levulinic acid−L-menthol in 1:1 molar ratio,<br>AT: CPC−UV | Out of various non-aqueous quaternary DES-based solvent systems developed for analysis of hydrophobic substances by CPC, two biphasic compositions such as i) a mixture of levulinic acid, L-menthol, methanol and *n*-heptane in 2.1:2.9:45:50 by weight and ii) a mixture of levulinic acid, L-menthol, methanol and *n*-heptane in 64:8.6:35:50 by weight were found the most promising for use as stationary as well as mobile phase in CPC separations. It was found that L-menthol does not precipitate out from a system containing low amount of DES. However, with high content of DES, water creates precipitation with L-menthol. Therefore, contact with water should be avoided [78] |

**Table 9** Applications of DESs in stationary phase as the preparative materials for different chromatographic techniques

| Analytes/Sample matrix | Experimental conditions | Remarks/reference |
|---|---|---|
| Ginsenosides, nucleosides and sulfonamides/N | SP: Sil-N-Glu<br>MP: A mixture of acetonitrile (85%) in water<br>DES composition: urea−ChCl, 2:1<br>AT: HPLC | This is the first paper reported by *Gu* et al. on the use of DESs in preparation of novel silica stationary phase (Sil-N-Glu) through silanization reaction using methylglucamine as functionalizing agent for use as HPLC column. The glucose-based silica material (Sil-N-Glu) exhibited remarkable separation efficiency associated with excellent selectivity for resolution of sulfonamides, nucleosides and ginsenosides [83] |
| Nucleosides/cordyceps | SP: Sil-NCDs<br>MP: A mixture of acetonitrile and ammonium acetate (20 mM) in 9:1 by volume (pH 6.6)<br>DES composition: Gly−ChCl, 3:1<br>AT: HILIC | The prepared Sil-NCDs (DES) column used as stationary phase in hydrophilic interaction liquid chromatography was found most effective for detection of nucleosides in cordyceps sample. The use of the developed Sil-NCDs phase broadened the application of carbon dots in designing new nano-composites in DESs. The proposed stationary phase has excellent potential for applications in the fields of pharmaceutical analysis [84] |
| Saccharides, amino acids and nucleosides | SP: Sil-PAA, Sil-PIm-PAA and Sil-PIm<br>MP: Acetonitrile + ammonium acetate (20 mM) in 9:1 by volume with pH 6.6<br>DES composition: Urea−ChCl (2:1),<br>EG−ChCl (2:1), Gly−ChCl (2:1)<br>AT: HILIC | The first reported use of DESs in the fabrication of Sil-PIm-PAA, Sil-PAA and Sil-PIm as silica-based polymer grafted stationary phases by radical chain transfer surface reaction. These stationary phases were used for successful separations of biomolecules (nucleosides, nucleobases, saccharides and amino acids) by HILIC. Sil-PAA and Sil-PIm-PAA demonstrated better selectivity as a result of stronger retention tendency for polar compounds. Furthermore, this study opened new avenues in the field of development of highly useful micro/nanomaterials for application in separation and related other research fields [85] |
| Nucleosides, nucleobases, saccharides, amino acids and ginsenosides/solution | SP: Sil-PIA<br>MP: A mixture of acetonitrile and ammonium acetate (50 mM) in 9:1 by volume (pH 6.6)<br>DES composition: EG−ChCl (3:1)<br>AT: HILIC | Using DESs, silica stationary phase grafted with itaconic acid was produced through surface chain transfer mechanism. Sil-PIA in combination with itaconic acid gave superior results in terms of selectivity and efficiency in comparisonof Sil-PIA + acylic acid. Further, Sil-PIA displayed excellent performance in the analysis and ameliorate separations of ginsenosides [86] |
| Nucleosides, nucleobases and ginsenosides/solution | SP: Sil-PEI, Sil-PEICDs<br>MP: A mixture of acetonitrile and ammonium acetate (50 mM) in 9:1 by volume (pH 6.6)<br>DES composition: EG−ChCl (3:1)<br>AT: HILIC | Because of favorable positive synergistic effect inside co-immobilized stationary phase, Sil-PEI/CDs showed outstanding separation selectivity for polar compounds due to manifest hydrogen-bonding interactions and adequate retention capability. In addition, Sil-PEI/CDs accomplished superior column performance as compared to Silica-PEI phase [87] |

**Table 9** (continued)

| Analytes/Sample matrix | Experimental conditions | Remarks/reference |
|---|---|---|
| Saccharides, nucleosides, nucleobases and amino acids /solution | SP: Sil-PDM-PIA, Sil-PDM-PAA<br>MP: A mixture of acetonitrile and ammonium acetate (50 mM) in 9:1 by volume (pH 2.6−2.98)<br>DES composition: EG−ChCl (3:1)<br>AT: HILIC | Two co-polymer grafted silica based stationary phases were synthesized in the presence of DESs through utilization of thiol-ene-click reaction. Out of four developed stationary phases such as Sil-PDM-PIA (S1), Sil-PDM-PAA (S2), Sil-PAA (S3) and Sil-PIA (S4) used in HILIC for the analysis of polar analytes, S1 and S2 stationary phases displayed enhanced retention and perceivable selectivity in comparison to S3 and S4 stationary phases probably due to the different properties of monomers. Furthermore, S1 and S2 columns seem to have latent potential for use in the analysis of medicinal plants extracts for identifying useful organic molecules of pharmaceutical importance with preliminary separation by HILIC [88] |

*DES* deep eutectic solvents, *HBD* hydrogen-bond donor, *HBA* hydrogen-bond acceptor, *EG* ethylene glycol, *Gly* glycerol, *ChCl* choline chloride, *BA* butyric acid/butanoic acid, *PA* pentanoic acid, *HA* hexanoic acid, *OA* octanoic acid, *DA* decanoic acid, *FA* formamide, *SP* stationary phase, *MP* mobile phase, *AT* analytical technique, *TLC* thin-layer chromatography, *HPTLC* high-performance thin-layer chromatography, *NADES* natural deep eutectic solvents, *HPLC* high-performance liquid chromatography, *ACN* acetonitrile, *QAs* quaternary alkaloids, *MTF* metformin, *MTP* metaprolol, *ATV* atorvastatin, *ASA* asprin, *SDS* sodium dodecyl sulphate, *MLC* micellar liquid chromatography, *MEL* melamine, *CPC* centrifugal partition chromatography, *CCC* counter-current chromatography, *CPE* centrifugal partition extractor, *LLC* liquid−liquid chromatography, *CBD* choline−1,4-butanediol, *BD* 1,4-butandiol, *CD* cyclodextrin, *NCDs* N-doped carbon dots, *Sil-N-Glu N*-methyl-glucamine-modified silica, *Sil-PAA* poly(acrylic acid)-grafted silica, *Sil-PIA* poly(itaconic acid)-grafted silica, *Sil-Plm* poly(1-vinylimidazole)-grafted silica, *Sil-PIM-PAA* poly(1-vinylimidazole-co-acrylic acid)-grafted silica, *HILIC* hydrophilic interaction chromatography, *DAD* diode-array detection, *UV* ultraviolet, *Vis* visible, *M.W.* molecular weight, *B.P.* boiling point, $t_r$ retention time, *Ip* interaction coefficient, $R_F$ ($k$) retention factor, *NA* not available

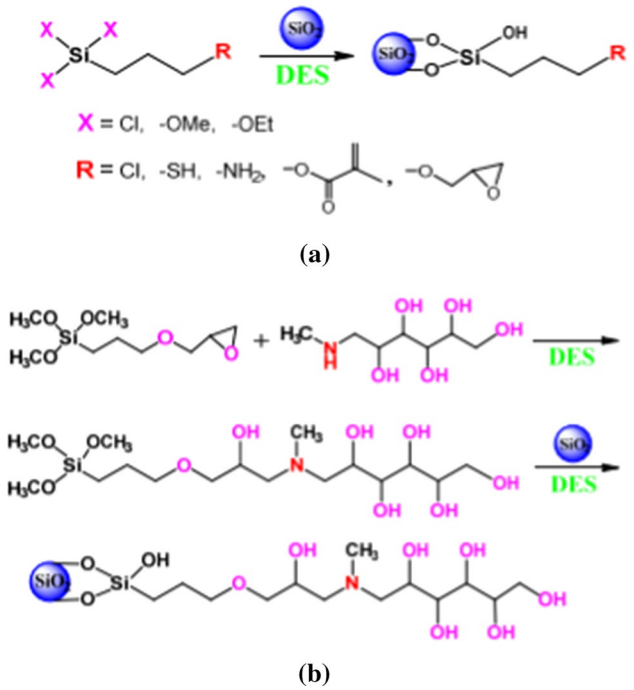



**Fig. 8** Use of DES media in silylation reactions (**a**). Steps involved in synthesis of spherical silica stationary phase modified with glucamine (**b**) [83]

(CCC/CPC) in 2016. This group used DESs as the substitute of water in biphasic system (a mixture of water and organic solvents). The most frequently used biphasic systems in CCC and CPC are composed of 3–4 solvents of different polarities [79–82]. DES-based biphasic systems attributed more choices to researchers for the separation of both hydrophobic and the hydrophilic compounds having low polarity. Several other studies have appeared in the literature after the first report of *Garcia* and his co-workers [68]. Table 8 describes the work done on the use of DESs in CCC/CPC since its first report [76–78].

The work reported in the literature on the applications of DESs in various liquid chromatographic techniques since their inception till 2021 is comprehensively summarized in Tables 7, 8, 9. It is clear from these tables that DESs have enormous possibilities for the use in synthetic and analytical chemical area of research. We foresee that in future several latent potentials of DESs will be uncovered, especially in the field of analytical science. Figures 7, 8, 9, 10, 11, 12 and Scheme 1 illustrate the various routes involved in the preparation of new stationary phases using in different chromatographic techniques.

## 5 Conclusion

This review is basically concerned with the compilation of scattered literature on the application of DESs in various modes of chromatographic techniques as the mobile phase additive/mobile phase, stationary phase component and as starting material for the preparation of stationary phase. From the literature survey, it may be concluded that traditional solvents which are volatile, hazardous and non-eco-friendly are being replaced by newer green solvents, particularly DESs or ionic liquids as sustainable liquid media

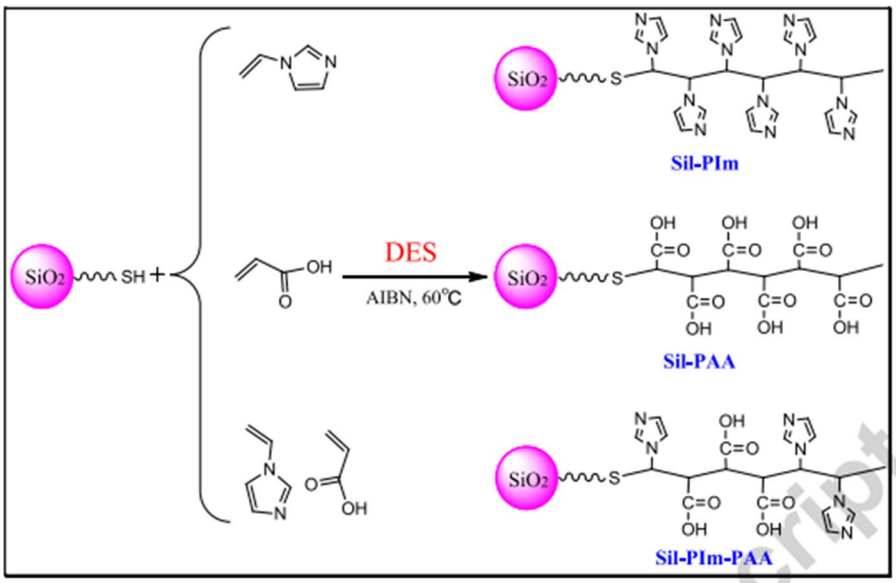



**Fig. 9** Preparation of silica-grafted stationary phases via radical chain–transfer surface reaction in the presence of DESs [85]

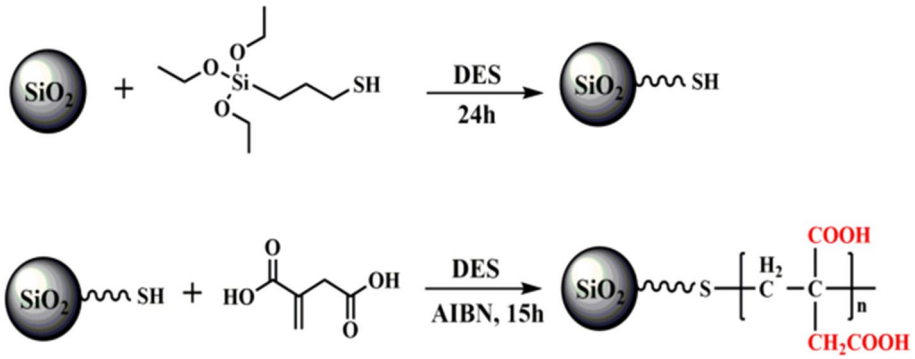



**Fig. 10** Preparation of poly(itaconic acid)-grafted silica stationary phase in the presence of DESs [86]

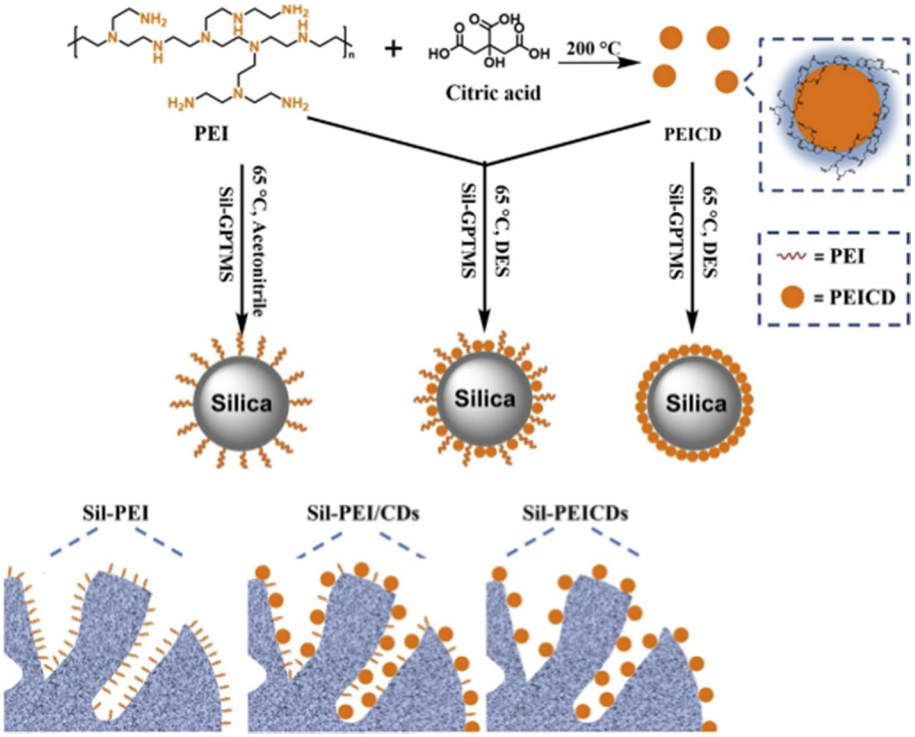



**Fig. 11** Illustration of co-immobilization of carbon dots functionalized polyethyleneimine and their precursor on silica surface [87]

in synthetic and analytical chemical sciences. As regards to chromatography, several combinations of hydrogen-bond donors and hydrogen-bond acceptors have been utilized to develop effective DESs for transforming chromatography into green chromatography. In this direction, substances of natural origin like amino acids and carbohydrates being economic and pollution-free have created renewed interest in researchers to explore more work on the use of DESs in chromatography. In DESs formation, ChCl has been more commonly used as HBA in combination of organic acids, phenol derivatives, urea or glycerol as HBD. Further, the literature survey also reveals that only one paper in the case of SFC and GC each has appeared on the use of DESs which can be a guiding source for researchers to explore the hidden possibilities of chromatographic potentiality of DESs. In fact, current prospects without the development of low viscosity DESs are limited except perhaps for GC and CCC. Furthermore, natural DESs being biodegradable have opened a new field to use them in chromatography of pharmaceutical products/drugs. However, certain issues related to high viscosity, ionic conductivity and water sensitivity have to be solved to expand the possibility of

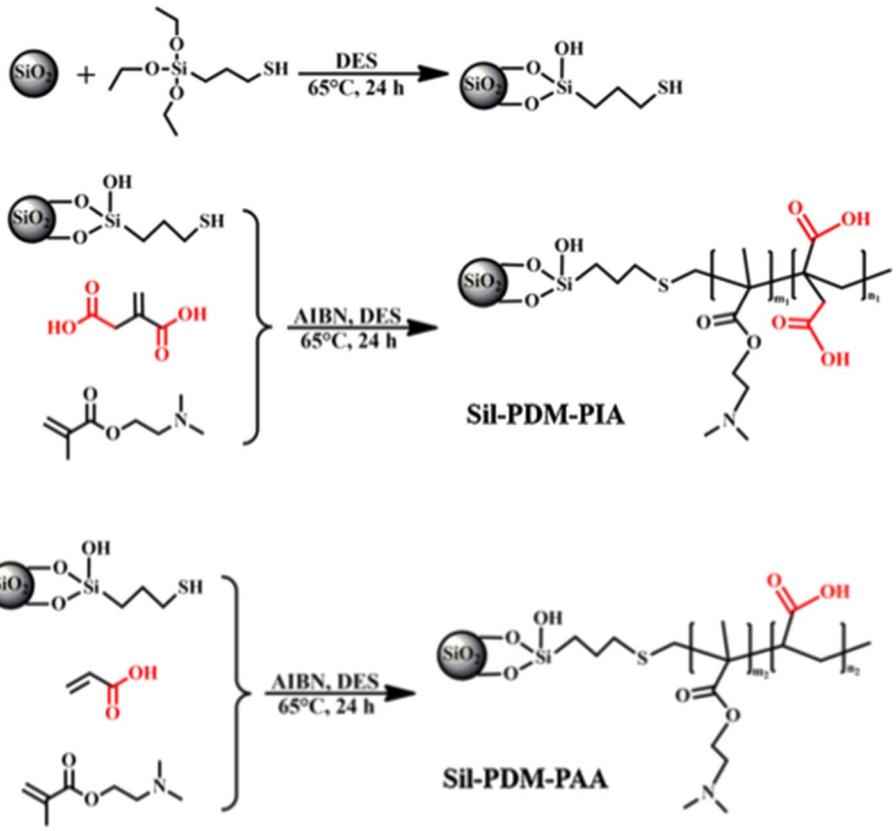



**Fig. 12** Structural representation of copolymer grafted silica stationary phases produced through surface thiol-en click reaction in DESs [88]




**Scheme 1** Development of carbon dot-enriched silica stationary phase with the use of deep eutectic solvents [84]

designing DESs–water combinations for desired applications. Since the use of DESs as mobile phase is not successful, they have been generally utilized as mobile phase additives. Though the use of DESs in the field of analytical chemistry is presently in its infancy, an increasing number of publications is expected in the near future.

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## Declarations

**Conflict of interest** The authors declare that they have no conflict of interest.

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